RTL-to-GDSII Design and Implementation of a 4-Bit Registered Array Multiplier in SkyWater 130 nm CMOS

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Digital VLSI Design — Assignment 1 (IIC-OSIC-TOOLS / `sky130_fd_sc_hd` Standard-Cell Library)

1. Introduction

Binary multiplication is a foundational arithmetic operation in digital signal processors, general-purpose microprocessors, and machine-learning accelerators. This report presents the complete end-to-end RTL-to-GDSII implementation of a 4-bit unsigned registered array multiplier (`mult_array`) in the open-source SkyWater 130 nm high-density standard-cell technology (`sky130_fd_sc_hd`, nominal corner `tt_025C_1v80` at V_{DD} = 1.80 V, T = 25 °C).

Given two 4-bit unsigned operands $A = a_3a_2a_1a_0 \in [0, 15]$ and $B = b_3b_2b_1b_0 \in [0, 15]$, the multiplier computes the 8-bit unsigned product $P = p_7p_6p_5p_4p_3p_2p_1p_0 = A \times B \in [0, 225]$ according to:

$$P = \sum_{i=0..3} \sum_{j=0..3} (a_j \cdot b_i)\, 2^{i+j}$$

The combinational core consists of 16 two-input AND gates that generate the 4 × 4 partial-product matrix $pp_{i,j} = a_j \wedge b_i$, followed by three ripple-carry addition stages (`Row 1`, `Row 2`, and `Row 3`) containing 4 full adders (`full_adder`) each (12 full adders in total). To support synchronous register-to-register static timing analysis (STA), the combinational multiplier array is bracketed by 8 input D flip-flops (`a_reg[3:0]`, `b_reg[3:0]`) and 8 output D flip-flops (`p[7:0]`) with synchronous active-low reset (`rst_n`), creating a deterministic 2-clock-cycle pipeline latency.

2. Design and Verification (Part A)

2.1 Architecture and Structural Hierarchy

The RTL implementation is partitioned into two synthesizable Verilog-2001 modules:

- `rtl/full_adder.v`: A 1-bit combinational full adder computing `sum = a ^ b ^ cin` and `cout = (a & b) | (b & cin) | (cin & a)`.
- `rtl/mult_array.v`: The top-level registered 4-bit array multiplier containing input registers `a_reg[3:0]` and `b_reg[3:0]`, four 4-bit partial-product vectors `pp0`–`pp3` (16 AND gates), 12 structural `full_adder` instances across three ripple-carry stages (`fa_r1_0..3`, `fa_r2_0..3`, `fa_r3_0..3`), and the 8-bit registered output `p[7:0]`.

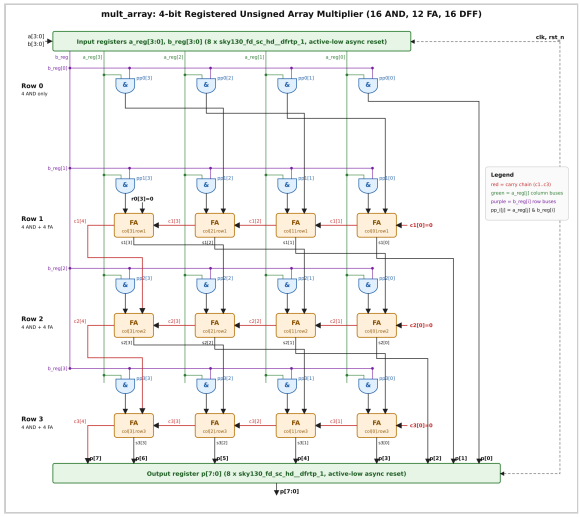


Figure 1: Hardware block diagram of `mult_array` showing the 8-bit input register bank (`a_reg[3:0]`, `b_reg[3:0]`), 16 partial-product AND gates, 12 full adders across three ripple-carry rows (`Row 1`–`Row 3`), and 8-bit output register (`p[7:0]`).

2.2 Hand-Worked Example for Roll Number `2023112024` (A = 9, B = 7)

For Roll Number `2023112024`, the last digit is d = 4, giving A = 4 + 5 = 9 = `4'b1001` and B = 7 = `4'b0111`, with expected product P = 9 × 7 = 63 = `8'b0011_1111` (`0x3F`). The four partial-product rows are `pp0 = A & {4{b[0]}} = 4'b1001`, `pp1 = A & {4{b[1]}} = 4'b1001`, `pp2 = A & {4{b[2]}} = 4'b1001`, and `pp3 = A & {4{b[3]}} = 4'b0000`. Tracing every internal signal in `rtl/mult_array.v`:

- **Direct LSB (Bit 0):** `p_comb[0] = pp0[0] = a[0] & b[0] = 1 & 1 = 1`.

- **Row 1** (adds `{1'b0, pp0[3:1]} = 4'b0100` and `pp1[3:0] = 4'b1001`): `fa_r1_0(a=0, b=1, cin=0)` → `s1[0]=1, c1[0]=0`; `fa_r1_1(a=0, b=0, cin=0)` → `s1[1]=0, c1[1]=0`; `fa_r1_2(a=1, b=0, cin=0)` → `s1[2]=1, c1[2]=0`; `fa_r1_3(a=0, b=1, cin=0)` → `s1[3]=1, c1[3]=0`. Thus `p_comb[1] = s1[0] = 1` and `{c1[3], s1[3:1]} = 4'b0110`.
- **Row 2** (adds `{c1[3], s1[3:1]} = 4'b0110` and `pp2[3:0] = 4'b1001`): `fa_r2_0(a=0, b=1, cin=0)` → `s2[0]=1, c2[0]=0`; `fa_r2_1(a=1, b=0, cin=0)` → `s2[1]=1, c2[1]=0`; `fa_r2_2(a=1, b=0, cin=0)` → `s2[2]=1, c2[2]=0`; `fa_r2_3(a=0, b=1, cin=0)` → `s2[3]=1, c2[3]=0`. Thus `p_comb[2] = s2[0] = 1` and `{c2[3], s2[3:1]} = 4'b0111`.
- **Row 3** (adds `{c2[3], s2[3:1]} = 4'b0111` and `pp3[3:0] = 4'b0000`): `fa_r3_0(a=1, b=0, cin=0)` → `s3[0]=1, c3[0]=0`; `fa_r3_1(a=1, b=0, cin=0)` → `s3[1]=1, c3[1]=0`; `fa_r3_2(a=1, b=0, cin=0)` → `s3[2]=1, c3[2]=0`; `fa_r3_3(a=0, b=0, cin=0)` → `s3[3]=0, c3[3]=0`. Thus `p_comb[7:3] = {c3[3], s3[3:0]} = 5'b00111`.

Combining all 8 product bits yields `p_comb[7:0] = 8'b0011_1111 = 63 = 0x3F`, matching the binary shift-and-add result ($9 \times 1 + 9 \times 2 + 9 \times 4 + 9 \times 0 = 9 + 18 + 36 + 0 = 63$).

2.3 Exhaustive Verification and Waveform Analysis

The self-checking testbench (`tb/tb_mult.v`) drives a 100 MHz clock (T_{clk} = 10.0 ns), holds `rst_n = 0` for 3 cycles, and exhaustively tests all 16 × 16 = 256 input combinations $(a, b) \in [0, 15]^2$. Each stimulus pair is applied on a `negedge clk`, followed by two `posedge clk` edges before comparing `p` against `expected = a * b`. Behavioral simulation with Icarus Verilog (`iverilog -o sim rtl/full_adder.v rtl/mult_array.v tb/tb_mult.v && vvp sim`) finishes with `PASS: 256 vectors checked, 0 errors`. Injecting a deliberate bug in `rtl/full_adder.v` (`assign sum = a ^ b;`, omitting `^ cin`) yields `FAIL: 256 vectors checked, 95 errors`, verifying the self-checking testbench.

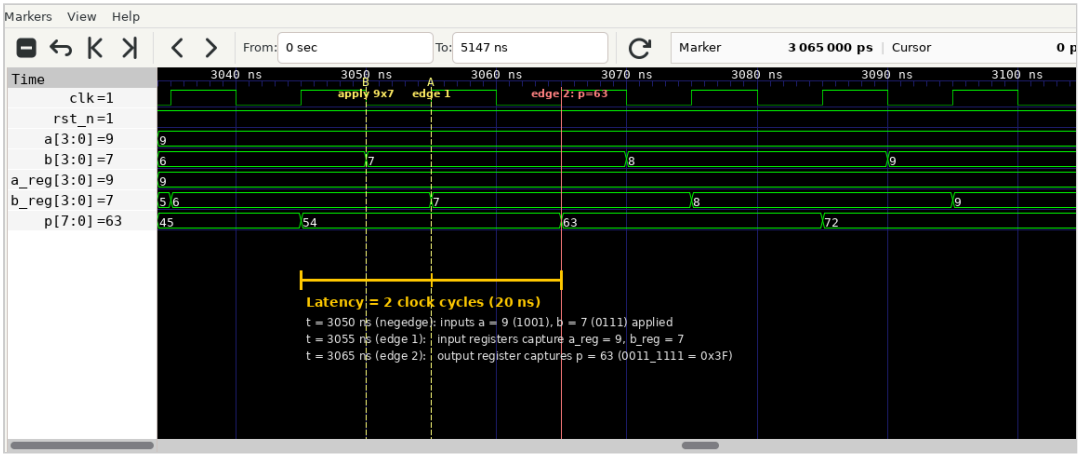


Figure 2: GTKWave simulation waveform demonstrating the 2-clock-cycle (20.0 ns) pipeline latency for the roll-number example $A = 9$, $B = 7$ → $P = 63$ (`0x3F`): inputs applied at $t = 3050$ ns, captured into `a_reg`/`b_reg` at Rising Edge 1 ($t = 3055$ ns), and latched into output `p[7:0]` at Rising Edge 2 ($t = 3065$ ns).

3. Logic Synthesis (Part B)

Logic synthesis was performed with Yosys 0.63 (`yosys -c synth/synth.tcl`) targeting the nominal SkyWater 130 nm standard-cell library (`sky130_fd_sc_hd__tt_025C_1v80.lib`). After hierarchy elaboration, process conversion (`proc`), design flattening (`flatten`), flip-flop mapping (`dfflibmap`), and ABC technology mapping (`abc -D 10000`), Yosys produced the gate-level netlist `synth/mult_array_netlist.v`. Gate-level simulation (GLS) with the Sky130 behavioral cell models confirmed 100% functional equivalence with `PASS: 256 vectors checked, 0 errors`. Tables 1 and 2 report the synthesis summary and complete 23-cell-type breakdown.

Table 1: Part B Logic Synthesis Summary for `mult_array` (`sky130_fd_sc_hd__tt_025C_1v80`).

Metric	Reported Value (with Units)
Total standard cell count	**67 cells** (51 combinational gates + 16 sequential flip-flops)
Flip-flop cell count (`sky130_fd_sc_hd__dfrtp_1`)	**16 cells** (8 input registers `a_reg`/`b_reg` + 8 output registers `p`)
Total chip area (cell area)	**746.966 µm^2**
Sequential (flip-flop) area & percentage	**400.384 µm^2** (**53.60%** of total cell area; 25.024 µm^2 per DFF)
Combinational logic area & percentage	**346.582 µm^2** (**46.40%** of total cell area; 51 logic gates)
Top 3 most frequent cell types	`dfrtp_1` (16 cells), `nand2_1` (12 cells), `xnor2_1` (7 cells)

Table 2: Complete 23-Cell-Type Breakdown of Synthesized `mult_array` from `synth/mult_array_stat.rpt`.

Cell Type (`sky130_fd_sc_hd__*`)	Count	Area (µm^2)	Cell Type (`sky130_fd_sc_hd__*`)	Count	Area (µm^2)
`dfrtp_1` (Pos-edge DFF w/ async rst_b)	16	400.384	`a22o_1` (2-2 AND-OR)	1	7.507
`nand2_1` (2-input NAND)	12	45.043	`a31oi_1` (3-1 AND-OR-Invert)	1	6.256
`xnor2_1` (2-input XNOR)	7	61.309	`a32oi_1` (3-2 AND-OR-Invert)	1	7.507
`maj3_1` (3-input Majority carry cell)	6	52.550	`and2_0` (2-input AND)	1	5.005
`nor2_1` (2-input NOR)	4	15.014	`and3_1` (3-input AND)	1	6.256
`xor2_1` (2-input XOR)	3	26.275	`and4_1` (4-input AND)	1	7.507
`a21oi_1` (2-1 AND-OR-Invert)	2	10.010	`inv_1` (Inverter)	1	3.754
`o211a_1` (2-1-1 OR-AND)	2	15.014	`o211ai_1` (2-1-1 OR-AND-Invert)	1	6.256
`o21a_1` (2-1 OR-AND)	2	12.512	`o21ai_1` (2-1 OR-AND-Invert)	1	5.005
`a211o_1` (2-1-1 AND-OR)	1	7.507	`o21ba_1` (2-1 OR-AND w/ inv B1_N)	1	6.256
`a21bo_1` (2-1 AND-OR w/ inv B1_N)	1	7.507	`or3_1` (3-input OR)	1	6.256
`xnor3_1` (3-input XNOR)	1	26.275	**Total (23 cell types)**	**67**	**746.966**

4. Static Timing Analysis (Part C)

Pre-layout Static Timing Analysis was executed in OpenSTA 2.7.0 (`sta sta/sta.tcl`) using `sky130_fd_sc_hd__tt_025C_1v80.lib`, 0.2 ns input/output delays relative to `clk`, and ideal interconnect (zero wire capacitance). At the baseline 10.0 ns clock period (100 MHz), setup WNS is **+7.396 ns** (reported as `+7.40 ns` at 2 decimal places) and hold WNS is **+0.264 ns** (`+0.26 ns`). Subtracting the setup slack yields the minimum clock period T_{min} = 10.00 − 7.396 = **2.604 ns** (**2.60 ns** to 0.01 ns resolution), corresponding to a pre-layout maximum clock frequency of F_{max} = 1 / 2.604 ns = **384.02 MHz** (or **384.62 MHz** at 2.60 ns). Tables 3 and 4 summarize the STA results and trace every gate along the critical path.

Table 3: Pre-Layout Static Timing Analysis Summary (`nom_tt_025C_1v80`).

Timing Metric	Reported Value (with Units)
Setup Worst Negative Slack (WNS) at T_{clk} = 10.00 ns	**+7.396 ns** (`+7.40 ns` at 2 decimals, `MET`)
Hold Worst Negative Slack (WNS) at T_{clk} = 10.00 ns	**+0.264 ns** (`+0.26 ns` at 2 decimals, `MET`)
Minimum clock period T_{min}	**2.604 ns** exact / **2.60 ns** (WNS = −0.004 ns / `-0.00 ns MET` at 2.60 ns)
Maximum pre-layout operating frequency F_{max} = 1 / T_{min}	**384.02 MHz** (at 2.604 ns) / **384.62 MHz** (at 2.60 ns)
Critical path startpoint → endpoint	`_106_` (`b_reg[0]`, `dfrtp_1`) → `_100_` (`p[6]`, `dfrtp_1`)
Data arrival time / Library setup time	**2.501 ns** arrival (0.340 ns CLK→Q + 2.161 ns comb.) + **0.103 ns** setup = **2.604 ns**

Table 4: Cell-by-Cell Critical Path Trace from `sta/mult_array_timing.rpt` (T_{clk} = 2.60 ns, `b_reg[0] → p[6]`).

#	Instance	Cell Type (Arc)	Output Net	Delay (ns)	Arrival (ns)	Edge	Architectural Stage in Array Multiplier
1	`_106_`	`dfrtp_1` (`CLK→Q`)	`b_reg[0]`	0.340	0.340	↑	Input register launching multiplier LSB b_0
2	`_043_`	`and2_0` (`A→X`)	`_009_`	0.175	0.515	↑	Partial-product AND gate: $pp_0[1] = b_0 \cdot a_1$
3	`_056_`	`or3_1` (`B→X`)	`_022_`	0.258	0.773	↑	Row 1 (`fa_r1_0` carry + `fa_r1_1`) intermediate OR term
4	`_058_`	`a32oi_1` (`A3→Y`)	`_024_`	0.148	0.921	↓	Row 1 (`fa_r1_1`) sum bit $s_1[1]$ (feeds Row 2 bit 0)
5	`_072_`	`o21ba_1` (`B1_N→X`)	`_038_`	0.214	1.135	↓	Row 2 (`fa_r2_0` carry → `fa_r2_1`) intermediate term
6	`_073_`	`xnor2_1` (`B→Y`)	`_039_`	0.314	1.448	↑	Row 2 (`fa_r2_1`) sum bit $s_2[1]$ (feeds Row 3 bit 0)
7	`_082_`	`maj3_1` (`B→X`)	`_048_`	0.273	1.721	↑	Row 3 (`fa_r3_0`) carry-out $c_3[0]$
8	`_084_`	`maj3_1` (`C→X`)	`_050_`	0.264	1.985	↑	Row 3 (`fa_r3_1`) carry-out $c_3[1]$
9	`_086_`	`xnor3_1` (`C→X`)	`p_comb[6]`	0.516	2.501	↑	Row 3 (`fa_r3_2`) sum $s_3[2]$ = `p_comb[6]`
10	`_100_`	`dfrtp_1` (setup `D`)	`p[6]`	0.103	2.604	—	Output register capturing product bit `p[6]`

5. Physical Design — Place and Route (Part D)

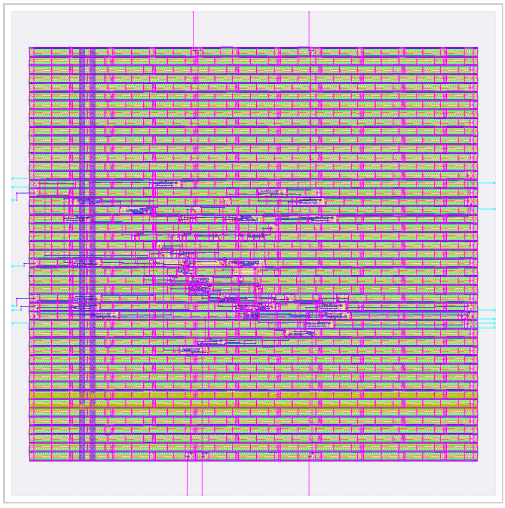
Automated RTL-to-GDSII physical implementation was performed using LibreLane 3.0.0.dev44 / OpenROAD on a fixed 150 µm × 150 µm die (22,500 µm^2) with a 130 µm × 130 µm core area (16,900 µm^2). Starting from the assignment guideline $T_{start} = \lceil 1.2 \times T_{min} \rceil_{0.5\,ns}$ = 3.5 ns, the target clock period was increased in 0.5 ns increments until post-route multi-corner STA achieved positive setup and hold slack across all 9 PVT corners, including the worst-case slow-slow corner (`max_ss_100C_1v60` at V_{DD} = 1.60 V, T = 100 °C). Table 5 records all 8 PnR runs, and Table 6 summarizes the final signoff metrics at T_{clk} = 7.0 ns.

Table 5: Iterative Clock-Period Sweep in LibreLane (3.50 ns to 7.00 ns in 0.50 ns increments).

Run	T_{clk} (ns)	Setup WNS (`nom_tt`)	Setup WNS (`max_ss`)	Hold WNS (worst)	Wirelength (µm)	Total Cells	DRC / LVS / Ant.	Signoff Status
1	3.50	+0.340 ns	−3.034 ns	+0.114 ns	2,324	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
2	4.00	+0.839 ns	−2.535 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
3	4.50	+1.339 ns	−2.035 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
4	5.00	+1.839 ns	−1.535 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
5	5.50	+2.339 ns	−1.035 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
6	6.00	+2.839 ns	−0.535 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
7	6.50	+3.339 ns	−0.035 ns	+0.113 ns	2,328	1,883	0 / 0 / 0	Setup Violation (`max_ss`)
8	**7.00**	**+3.839 ns**	**+0.465 ns**	**+0.113 ns**	**2,328**	**1,883**	**0 / 0 / 0**	**PASS (All 9 Corners)**

Table 6: Final Post-Route Physical Design Metrics at Signoff (T_{clk} = 7.00 ns, `pnr/metrics.json`).

Physical / Timing Metric	Value (with Units)	Physical / Timing Metric	Value (with Units)
Die area (150 µm × 150 µm)	**22,500.00 µm^2**	Core area (130 µm × 130 µm)	**16,900.00 µm^2**
Active standard-cell area	**1,316.01 µm^2**	Core utilization	**7.787%**
Active instances (excl. fill/decap)	**319 cells**	Total physical cells (incl. fill/decap)	**1,883 cells**
Clock buffers / Timing repair buffers	**2 / 11 buffers**	Tap cells / Decap + fill cells	**239 / 1,564 cells**
Total routed wirelength	**2,328 µm**	Total routed vias	**517 vias**
Post-route setup WNS (`max_ss_100C_1v60`)	**+0.465 ns**	Post-route setup WNS (`nom_tt_025C_1v80`)	**+3.839 ns**
Post-route hold WNS (worst corner)	**+0.113 ns**	Worst-case clock skew	**0.017 ns**
Post-route F_{max} (`max_ss_100C_1v60`)	**153.01 MHz**	Post-route F_{max} (`nom_tt_025C_1v80`)	**316.34 MHz**
Total power (`nom_tt` at 7.0 ns)	**0.244 mW**	Magic DRC / Netgen LVS / Antenna	**0 / 0 / 0 violations**



(a) Full-die view (150 µm × 150 µm): Power distribution grid (`met4`/`met5` straps), perimeter I/O pins (`met2`/`met3`), standard-cell placement rows, and decap/filler array.



(b) Zoomed active logic region (35 µm × 30 µm): Placed `dfrtp_1`, `maj3_1`, and `xnor2_1` standard cells, local interconnect (`li1`), `met1`/`met2` tracks, and vias.

Figure 3: KLayout GDSII renders of the final DRC/LVS-clean `pnr/mult_array.gds` layout (Run 8 at T_{clk} = 7.0 ns).

6. Answers to Assignment Questions (Questions 1–6)

Question 1 (Area Breakdown — Sequential vs. Combinational Area):

From Yosys synthesis (`synth/mult_array_stat.rpt`), the 16 `sky130_fd_sc_hd__dfrtp_1` flip-flops consume **400.384 µm^2** out of the total cell area of **746.966 µm^2**, accounting for **53.60%** of the entire design area (25.024 µm^2 per DFF), whereas the 51 combinational cells implementing the entire 4 × 4 multiplier array occupy **346.582 µm^2** (**46.40%**). Flip-flops dominate because each master-slave edge-triggered D flip-flop with asynchronous reset requires two clocked latches, transmission gates, internal clock buffering, and reset logic (~24–28 transistors, 6.67× the area of a 3.754 µm^2 `nand2_1` gate). Furthermore, for an N-bit registered multiplier, sequential register area scales linearly as $O(N)$ ($4N$ flip-flops: $2N$ input + $2N$ output), whereas the combinational array scales quadratically as $O(N^2)$ (N^2 AND gates + $N(N-1)$ adders). At $N = 4$, $4N = 16$ flip-flops is numerically equal to $N^2 = 16$ partial products; for wider multipliers ($N = 16, 32, 64$), the $O(N^2)$ combinational array dominates.

Question 2 (Architectural Critical Path vs. Multiplier Topology):

As traced in Table 4, the OpenSTA worst-case setup path begins at `b_reg[0]` (`_106_`), traverses 1 partial-product AND gate (`_043_`, $pp_0[1] = b_0 \cdot a_1$) and 7 combinational full-adder cells spanning **7 full-adder stages** (`fa_r1_0` → `fa_r1_1` → `fa_r2_0` → `fa_r2_1` → `fa_r3_0` → `fa_r3_1` → `fa_r3_2`), and terminates at output flip-flop `p[6]` (`_100_`). This matches the theoretical longest path of an $N \times N$ Braun ripple-carry array multiplier, where partial products generated in the upper-right corner must ripple diagonally/vertically down all three adder rows and then horizontally along the final ripple-carry adder row (`Row 3`). Although `p[7]` is the MSB carry-out ($c_3[3]$), in `sky130_fd_sc_hd` the 3-input XNOR sum cell (`xnor3_1`, 0.516 ns delay) driving `p_comb[6]` is slower than the 3-input majority carry cell (`maj3_1`, ~0.26 ns delay), making `p[6]` the critical endpoint at 2.501 ns data arrival time.

Question 3 (Pre-Layout vs. Post-Route F_{max} Comparison):

Pre-layout STA in Part C reported T_{min} = 2.604 ns (F_{max} = **384.02 MHz**) at the nominal corner (`nom_tt_025C_1v80`) with zero

interconnect parasitics and ideal clock distribution. After physical place-and-route in Part D:

- **Same-Corner Comparison (`nom_tt_025C_1v80`):** At T_{clk} = 7.0 ns, post-route setup slack is +3.839 ns, giving $T_{min,post,nom}$ = 7.0 − 3.839 = 3.161 ns and $F_{max,post,nom}$ = **316.34 MHz** (a **17.6%** reduction). This slowdown comes from extracted RC interconnect parasitics across 2,328 µm of metal wiring and 517 vias, clock-tree skew (0.017 ns), and 11 inserted slew/hold buffers.
- **Multi-Corner Signoff Comparison (`max_ss_100C_1v60`):** Across all PVT corners, the slow-slow corner (1.60 V, 100 °C) limits setup slack to +0.465 ns at 7.0 ns, giving $T_{min,post,ss}$ = 6.535 ns and $F_{max,post,ss}$ = **153.01 MHz** (a **60.2%** reduction relative to pre-layout nominal STA due to reduced supply voltage and elevated temperature degrading MOSFET drive current).

Question 4 (Pre-Layout vs. Post-Route Cell Count — 67 vs. 1,883 Cells):

Yosys synthesis produced **67 cells** (746.966 µm^2). After PnR, the layout contains **319 active instances** (1,316.01 µm^2) and **1,883 total physical cells** once 1,564 decap/filler cells are included. The additional 1,816 physical cells consist of:

- **Substrate/Well Tap Cells (+239 cells, `tapvpwrvgnd_1`):** Placed periodically in every row to tie the p-substrate to `VGND` and n-well to `VPWR`, preventing CMOS latch-up.
- **Clock-Tree Buffers (+2 cells, `clkbuf_2`, `clkbuf_16`):** Inserted during CTS to drive the 16 flip-flop clock pins with low skew (0.017 ns).
- **Timing & Slew Repair Buffers (+11 cells, `buf_1`, `buf_2`, `clkbuf_1`):** Inserted by OpenROAD `RepairDesign`/`RepairTiming` to buffer I/O nets and satisfy max-slew and hold constraints.
- **Decoupling Capacitors & Non-Functional Fillers (+1,564 cells, `decap_*`, `fill_*`):** Fill all unoccupied row sites across the 16,900 µm^2 core (92.2% empty space at 7.79% utilization) to ensure continuous n-well and power-rail geometry for DRC compliance and local charge storage against dynamic IR drop.

Question 5 (Architectural Change for 2× Higher F_{max} in the Same Technology):

To achieve a 2× increase in F_{max} without changing the `sky130_fd_sc_hd` library:

1. **Internal Pipeline Stage Insertion:** Adding an intermediate pipeline register stage between `Row 1` and `Row 2` (or between `Row 2` and `Row 3`) partitions the 7-full-adder combinational path into two balanced stages of ≤4 full adders each. This cuts the combinational delay in half, nearly doubling F_{max} (from ~384 MHz to >700 MHz pre-layout) at the expense of +1 clock cycle latency (3 cycles total) and ~10 flip-flops.
2. **Wallace / Dadda Tree Multiplier with Carry-Lookahead Final Adder (CLA):** Replacing the sequential ripple-carry rows with parallel 3:2 carry-save compressors (Wallace/Dadda tree) followed by a fast 6-bit Carry-Lookahead or Kogge-Stone parallel-prefix adder reduces the critical path from 7 series full adders to 2 carry-save stages plus a logarithmic prefix adder (~4 gate stages), nearly halving combinational delay while preserving the 2-cycle latency.

Question 6 (Why the First Adder in Each Row Uses a Full Adder with `cin = 0`):

In `rtl/mult_array.v`, the least-significant adder of each row (`fa_r1_0`, `fa_r2_0`, `fa_r3_0`) adds only two operands and ties `.cin(1'b0)`. Algebraically, setting c_{in} = 0 in a full adder reduces the logic equations to `sum = a ^ b ^ 0 = a ^ b` and `cout = (a & b) | (b & 0) | (0 & a) = a & b`, which is functionally identical to a half adder. Instantiating `full_adder` with `cin = 1'b0` keeps the RTL hierarchy clean and uniform (requiring only one adder module definition instead of separate half-adder and full-adder modules) while incurring **zero area or delay penalty**: during synthesis, Yosys constant propagation (`opt_expr` and ABC logic optimization) automatically simplifies the constant-0 input and maps those three instances to 2-input XOR/AND/NAND gates rather than 3-input full-adder cells.

7. Conclusion

A 4-bit unsigned registered array multiplier (`mult_array`) was designed in Verilog-2001, exhaustively verified across all 256 input pairs in both behavioral and gate-level simulation, synthesized to 67 standard cells (746.966 µm^2) in `sky130_fd_sc_hd`, characterized with OpenSTA (T_{min} = 2.60 ns, F_{max} = 384.02 MHz), and implemented through LibreLane/OpenROAD place-and-route into a DRC-, LVS-, and antenna-clean GDSII layout achieving 153.01 MHz multi-corner signoff operation (316.34 MHz at the nominal corner).